MCC Entrance Test, August 2026 — Day 2 — Solutions

Level T2

Test Information. Level T2 — 120 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra, Combinatorics, Geometry (×2), Number Theory (×2).
Rules: show your work (partial credit rewards genuine progress); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts).[1] Two numbers x and y satisfy $x + y = 4$ and $xy = 3$. What is the value of $(x - y)^2$?

Official solution. Expand $(x - y)^2$ in terms of $(x + y)^2$ and xy: $(x - y)^2 = (x + y)^2 - 4xy = 4^2 - 4(3) = 16 - 12 = \boxed{4}$. □

Problem 2 (10 pts).[2] A rectangular box (a cuboid) has length 3, width 3, and height 23. A cube has the same total surface area as this box. Find the side length of the cube.

Official solution. The box has two 3×3 faces and four 3×23 faces, so its total surface area is

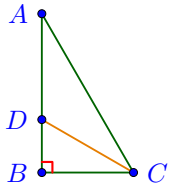
$$2(3 \cdot 3) + 4(3 \cdot 23) = 18 + 276 = 294.$$

A cube with side s has surface area $6s^2$, so $6s^2 = 294 \implies s^2 = 49 \implies s = \boxed{7}$. □

Problem 3 (15 pts).[3] In right triangle ABC, the right angle is at B. Point D lies on segment AB with $DB = 10$. Given that $\angle CDB = 60°$ and $\angle ACD = 30°$,

(a) Find the length of CD. [6 pts]

(b) Find the length of AD. [9 pts]



Official solution. **Part (a).** $\triangle CDB$ has a right angle at B and $\angle CDB = 60°$, so it is a 30-60-90 $\triangle$ with DB opposite the 30° angle at C as its shorter leg. Hence the hypotenuse CD is twice DB. $CD = 2 \times DB = \boxed{20}$.

Part (b). Since A, D, B are collinear with D between A and B, $\angle ADC = 180° - \angle CDB = 120°$. In triangle ADC, the angles sum to 180°, so $\angle DAC = 180° - 120° - 30° = 30°$. Since $\angle DAC = \angle ACD = 30°$, triangle ADC is isosceles with $AD = CD$, so $AD = CD = \boxed{20}$. □

Problem 4 (15 pts).[4] A trench has two rows of pits, one row along each side, with 10 pits in each row. A total of 7 diggers are to be placed into pits so that, within each row, no two diggers are in adjacent pits (a digger in one row may be next to a digger in the other row — only pits in the *same* row count as adjacent).

(a) In one row of 10 pits, find the number of ways to place k diggers with no two adjacent, for each $k = 0, 1, 2, 3, 4, 5$. [9 pts]

(b) Using part (a), find the number of ways to place all 7 diggers across the two rows, following the rule above. [6 pts]

Official solution. **Part (a).** Choosing k non-adjacent positions from a row of n is equivalent to choosing k positions from a row of $n - k + 1$ (*slide each chosen position left by the number of gaps already forced before it*), so the count is $\binom{n-k+1}{k}$.

$$\underbrace{- * - * - - * - * -}_{\text{choose 4 from 10}} \quad \to \quad \underbrace{- * - * - * - * -}_{\text{choose 4 from } 10-4+1=7}$$

For $n = 10$:

$$k = 0, 1, 2, 3, 4, 5 \longmapsto \binom{11}{0}, \binom{10}{1}, \binom{9}{2}, \binom{8}{3}, \binom{7}{4}, \binom{6}{5} = \boxed{1,\ 10,\ 36,\ 56,\ 35,\ 6}.$$

Note that no larger k is possible, since $2k > 11$ once $k \geq 6$.

[1] Hanoi Open Mathematics Competition. [2] IMO Malaysia National Screening Test (IMONST). [3] Euclid Contest (CEMC, University of Waterloo). [4] British Mathematical Olympiad, Round 1 (BMO1).

Part (b). If k diggers are placed in one row and $7-k$ in the other, *the two rows are filled independently, so the count for that split is the product of the two values* from part (a). Only $k = 2, 3, 4, 5$ give a nonzero product (since $k \geq 6$ or $7-k \geq 6$ is impossible):

$$(k, 7-k) = (2,5), (3,4), (4,3), (5,2) \implies 36{\cdot}6,\ 56{\cdot}35,\ 35{\cdot}56,\ 6{\cdot}36 = 216,\ 1960,\ 1960,\ 216.$$

Summing, $216 + 1960 + 1960 + 216 = \boxed{4352}$. □

Problem 5 (25 pts).[5] Consider the numbers $1, 2, 3, \ldots, 99$, and the alternating sum formed by adding and subtracting their squares in order, starting with a plus sign:

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots - 98^2 + 99^2 = \underbrace{(1^2 - 2^2)}_{k=1} + \underbrace{(3^2 - 4^2)}_{k=2} + \cdots + \underbrace{(97^2 - 98^2)}_{k=49} + 99^2.$$

(a) Show that the k-th paired term $(2k-1)^2 - (2k)^2$ simplifies to $-(4k-1)$. [9 pts]

(b) Using part (a), evaluate $\sum_{k=1}^{49} -(4k-1)$, the sum of all 49 paired terms. [3 pts]

(c) Add back the unpaired term 99^2 to find the value of the full alternating sum, then find the greatest prime number that divides it. [13 pts]

Official solution. **Part (a).** The k-th pair is $(2k-1)^2 - (2k)^2$, a difference of squares:

$$(2k-1)^2 - (2k)^2 = \big[(2k-1) - 2k\big]\big[(2k-1) + 2k\big] = (-1)(4k-1) = \boxed{-(4k-1)}.$$

Part (b). Summing the formula from part (a) over $k = 1$ to 49:

$$\sum_{k=1}^{49} -(4k-1) = -4\sum_{k=1}^{49} k + 49 = -4 \cdot \frac{49 \cdot 50}{2} + 49 = -4900 + 49 = \boxed{-4851}.$$

Part (c). Adding back the unpaired term $99^2 = 9801$:

$$-4851 + 9801 = 4950 = 2 \cdot 3^2 \cdot 5^2 \cdot 11, \text{ hence the greatest prime factor is } \boxed{11}.$$

□

Problem 6 (25 pts).[6] A two-digit number is called *lucky* if it is evenly divisible by the sum of its own digits. (For example, 12 is lucky since $1 + 2 = 3$ divides 12; but 13 is not, since $1 + 3 = 4$ does not divide 13.) How many lucky two-digit numbers are there?

(a) Write the number as $\overline{tu} = 10t + u$, with tens digit $t \in \{1, \ldots, 9\}$ and units digit $u \in \{0, \ldots, 9\}$. Show that $(t+u) \mid (10t+u)$ if and only if $(t+u) \mid 9t$. [7 pts]

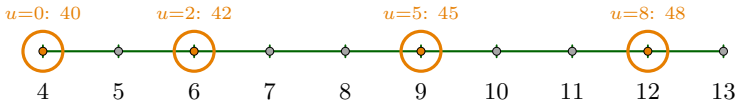
(b) Explain why, for a fixed tens digit t, this restricts $t+u$ to being one of the (finitely many) divisors of $9t$ that also lie in the range $[t, t+9]$ (the range $t+u$ can take as u runs over $0, \ldots, 9$). [3 pts]

(c) Using parts (a)–(b), check each tens digit $t = 1, \ldots, 9$ and count the total number of lucky two-digit numbers. [15 pts]

Official solution. **Part (a).** Since $10t + u = 9t + (t+u)$, the number $t+u$ divides $10t+u$ exactly when it divides $9t + (t+u) - (t+u) = 9t$ as well – i.e. $(t+u) \mid (10t+u) \iff (t+u) \mid 9t$, because $t+u$ trivially divides itself, so it divides $10t+u$ if and only if it divides the difference $9t$.

[5] Purple Comet! Math Meet, Middle School Division. [6] Singapore Mathematical Olympiad.

Part (b). As u ranges over $0, \ldots, 9$ with t fixed, $t+u$ ranges over the 10 consecutive integers $t, t+1, \ldots, t+9$. By part (a), a lucky number needs $t+u$ to be a divisor of $9t$, so only divisors of $9t$ that also fall in $[t, t+9]$ can occur – a short, checkable list for each t.



$t = 4$: divisors of $9t = 36$ in the range $[4, 13]$ are $4, 6, 9, 12$ (circled) – giving the 4 lucky numbers $40, 42, 45, 48$

Part (c). Checking each tens digit $t = 1, \ldots, 9$ against its range $[t, t+9]$ of possible values of $t+u$, and keeping only those that divide $9t$:

- $t = 1$: divisors of 9 in $[1, 10]$: $1, 3, 9 \to u = 0, 2, 8$ (numbers $10, 12, 18$)
- $t = 2$: divisors of 18 in $[2, 11]$: $2, 3, 6, 9 \to u = 0, 1, 4, 7$ (numbers $20, 21, 24, 27$)
- $t = 3$: divisors of 27 in $[3, 12]$: $3, 9 \to u = 0, 6$ (numbers $30, 36$)
- $t = 4$: divisors of 36 in $[4, 13]$: $4, 6, 9, 12 \to u = 0, 2, 5, 8$ (numbers $40, 42, 45, 48$)
- $t = 5$: divisors of 45 in $[5, 14]$: $5, 9 \to u = 0, 4$ (numbers $50, 54$)
- $t = 6$: divisors of 54 in $[6, 15]$: $6, 9 \to u = 0, 3$ (numbers $60, 63$)
- $t = 7$: divisors of 63 in $[7, 16]$: $7, 9 \to u = 0, 2$ (numbers $70, 72$)
- $t = 8$: divisors of 72 in $[8, 17]$: $8, 9, 12 \to u = 0, 1, 4$ (numbers $80, 81, 84$)
- $t = 9$: divisors of 81 in $[9, 18]$: $9 \to u = 0$ (number 90)

Totaling $3 + 4 + 2 + 4 + 2 + 2 + 2 + 3 + 1 = \boxed{23}$ lucky two-digit numbers. □